

Admission-Aware Constrained Reinforcement Learning for Semantic Evidence Scheduling in U-Space UAV Networks

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Abstract—Multi-unmanned-aerial-vehicle (UAV) visual question answering (VQA) in the low-altitude economy couples a semantic-communication decision—which *level* of evidence (a cached answer, symbolic detector tokens, or the image) to transmit for each question—with a safety-critical airspace resource-allocation problem regulated by U-space separation management. We make three contributions. First, we build, to our knowledge, the first U-space/BUBBLES-conformant semantic evidence-scheduling simulation framework in which every scenario parameter is traceable to a SESAR Joint Undertaking deliverable; the separation kernel reproduces the official G-4/G-5 numerics to within 0.18%. A corrected *safety bridge* couples the payload to the separation chain through an M/G/1 priority queue: token uploads leave the tactical-contact distance at the BUBBLES baseline (370.6 vs. 370.5 m), while image uploads at peak load inflate it by 23.7 m and forfeit $\approx 6\%$ of relative airspace capacity at -5 dB—an asymmetric airspace tax. Second, we formulate scheduling as a CMDP on the admitted set with a certificate-based escalation channel: a per-slot spec-attainability certificate adjudicates unserveable safety-critical queries into a budgeted escalation path (δ_{esc} calibrated from the certificate itself), and a risk-aware constrained hybrid-action PPO with per-channel dual caps and warm-up is trained on top. Third, the same dual machinery is operated across a feasible (nominal) and an infeasible (peak-overload) condition and exhibits both textbook regimes: under nominal the quality dual converges interior (3.29, never at its cap) and the policy reaches 93% of the escalation-aware oracle’s admitted mission success (0.653 vs. 0.703); under peak the dual ratchets to its cap on 3/3 seeds—the classical infeasibility signature—the constrained and unconstrained arms coincide, and safety is carried structurally: the escalation-aware oracle attains 0.916 only by escalating 0.666 of the critical load, while conflict, deadline-critical, and escalation budgets hold (0.003 vs. 0.08; 0.086 vs. 0.155). A per-channel accounting table, a flat-PPO baseline, dual-machinery ablations, four-axis zero-shot sweeps, and honest red lines (quality channels are priced, approached, but not closed) complete a reporting-standard evaluation.

Index Terms—Semantic communication, constrained reinforcement learning, U-space, UAV networks, resource allocation, visual question answering, safe RL, hybrid action space, Lagrangian methods.

I. INTRODUCTION

THE low-altitude economy is turning urban airspace into a shared resource for fleets of small unmanned aerial vehicles (UAVs) performing inspection, delivery, and surveillance. A recurring application is aerial *visual question answering* (VQA): a ground operator or an automated safety service asks natural-language questions about a live UAV feed (“how many vehicles at this junction?”, “is the pad clear?”), and an edge server must answer over a bandwidth-limited, error-prone air-to-ground (A2G) link. Reconstructing the full image for every question is wasteful; a companion study shows that most questions are answered more cheaply—and often more accurately—from lower *levels* of evidence: a cached answer, a few symbolic detector tokens, or the raw image [1]. That study characterizes the per-question evidence–question complementarity in isolation. The present paper asks the systems question that follows: when *many* UAVs and *many* questions contend for spectrum, compute, and—crucially—the same regulated airspace, how should evidence levels and physical resources be scheduled *jointly*, under hard safety constraints?

Two facts make this more than a standard resource-allocation problem. First, the airspace is governed. Under the European U-space framework [2], a separation management service keeps aircraft apart to a target level of safety; the SESAR JU BUBBLES project [3] formalizes the separation minima, the traffic performance envelope, and the demand profile that any conformant simulation must respect. Most learning-based UAV resource papers invent their own mobility, arrival, and conflict models, which makes their safety claims hard to audit. Second, the communication choice *is* a safety variable: the tactical-contact separation distance in BUBBLES grows with a communication-delay term, so transmitting the heavy image instead of light tokens over a weak link literally pushes aircraft apart and shrinks airspace capacity. Semantic communication thus enters the separation loop, not merely the throughput budget.

We turn these observations into a single simulation-and-control study with three contributions.

(1) **A U-space/BUBBLES-conformant semantic evidence-scheduling framework with a corrected safety bridge.** Every scenario parameter is anchored to a SESAR deliverable: the traffic performance envelope (cruise/climb speeds, airframe size), the Appendix-G demand profile (daily-mean 8.17 and peak 20 concurrent aircraft, giving a nominal and a peak operating condition), the chained tactical-contact separation

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minimum d_{TC} , and a per-slot conflict budget of ≈ 0.08 obtained by walking the $TLS_{MAC} = 2.5 \times 10^{-7}$ fatal-accidents-per-flight-hour figure up the mitigation-barrier ladder. Our separation kernel reproduces the official Table G-4/G-5 numerics to within 0.18% (a $370.53 \rightarrow 370.49$ m contact-distance check). On top of this we build a *safety bridge* whose coupling we state honestly: the payload contends with the separation service's own exchange on the shared RAN, modelled as an M/G/1 priority queue on top of the official 1.8 s separation-communication mean. Token uploads add milliseconds (separation stays at the BUBBLES baseline), while image uploads at peak load add 1.98 s of queueing wait, inflating d_{TC} by 23.7 m and forfeiting $\approx 6\%$ of relative airspace capacity at -5 dB—heavy evidence *taxes* the airspace; nothing beats the baseline.

(2) An admitted-set CMDP with a certificate-based escalation channel. Peak overload contains queries that no transmission service can serve within spec, so we make unserveability a certified state: a per-slot *spec-attainability certificate* (an environment property, recomputed from the remaining deadline) adjudicates every critical query that ends rejected or expired—serveable ones are charged to the policy, unserveable ones are *escalated* to a higher tier against a budget δ_{esc} that is itself calibrated from the certificate (0.155 peak, 0.05 nominal). The rate constraints are stated on the *admitted* (non-escalated) set, which is the promise an overloaded safety system can actually keep. On top we train a risk-aware constrained hybrid-action PPO (two-timescale actor, eight dual channels with per-channel caps, dual warm-up, and a six-layer constraint stack), audited by an epoch-0 policy-ratio $\equiv 1$ gate and a 321-test suite.

(3) A dual-pathology study across a feasible and an infeasible regime, with a reporting-standard evaluation matrix. Operating the same dual machinery in both conditions yields both textbook behaviours: under nominal the quality-critical multiplier converges *interior* (3.29, never touching its cap of 8) and the learned policy reaches 93% of the escalation-aware oracle's admitted mission success (0.653 vs. 0.703); under peak the same multiplier ratchets to its cap on 3/3 seeds—the classical CMDP infeasibility signature—the constrained and unconstrained arms coincide, and the correct safety story is architectural: the escalation-aware oracle attains admitted mission 0.916 only by escalating 0.666 of the critical load, while a non-escalating oracle collapses to 0.528. We report a per-channel constraint-accounting table (conflict 0.003 vs. budget 0.08; escalation 0.086 vs. 0.155; quality channels priced but not closed), a flat single-head PPO baseline, dual-machinery ablations (channel-off, warm-start-at-the-pin, no slow head), four-axis zero-shot generalization (UAV count, arrival load, link attenuation, interference floor) plus a dedicated three-point zero-shot section, and explicit red lines throughout.

The rest of the paper is organized as follows. Section II positions the work against semantic-aware DRL resource allocation, constrained/safe RL for wireless, and U-space separation management. Section III develops the BUBBLES-conformant system model, the M/G/1 safety-bridge coupling, the deadline QoS anchoring, and the spec-attainability cer-

tificate. Section IV formalizes the admitted-set CMDP with the escalation budget. Section V details the two-timescale network, the six-layer constraint machinery, the correctness fixes, and the training configuration. Section VI reports the evaluation matrix, the dual-pathology evidence, and the corrected safety bridge. Section VII distills the pricing-versus-admission lesson for constrained-RL practice, and Section VIII concludes.

II. RELATED WORK

We position the paper against three lines: semantic-aware DRL resource allocation, constrained/safe RL for wireless, and U-space separation management. Table-free boundaries are drawn explicitly at the end of each subsection.

A. Semantic-aware DRL resource allocation

Task-oriented and semantic communication has shifted the design objective from bit fidelity to downstream task success [4], [5], with VQA a canonical receiver task [6], [7]. Park and Yoon [8] transmit task-adaptive visual semantics—the closest single-link neighbour of our evidence levels—but on one link with no fleet contention, no airspace coupling, and no admission decision; adaptive resource allocation for semantic-communication networks is studied in [9], and hybrid bit/semantic transmission reaches UAV networks in [10], again without a safety regulator in the loop. When many users and links share scarce spectrum and compute, the allocation is naturally cast as a DRL problem. Ji *et al.* [11] use multi-agent PPO for semantic-aware allocation on a VQA-bearing workload, with an exhaustive upper bound and a 100-episode test protocol. Long *et al.*'s Lya-HiPPO [12] couples Lyapunov-guided constraint handling with hierarchical PPO for semantic-aware AoI minimization in UAV-assisted networks and is the source of our flat-PPO baseline convention. Hybrid discrete-continuous action spaces—selecting a mode and tuning its parameters—appear in parameterized-action RL [13] and its wireless adaptations [14], and value-based methods with conventional-optimization baselines are common in the IoT setting [15]. Our companion paper [1] establishes the per-question evidence-question complementarity that our scheduler exploits, but it performs no multi-agent scheduling and imposes no safety constraints. *Boundary*: unlike these works we (i) route among discrete *evidence levels* rather than modulation/rate/power alone, (ii) embed the allocation in an externally specified U-space safety scenario, and (iii) audit which mechanism actually carries the safety load—pricing or admission—rather than assuming the learned policy satisfies the constraints.

B. Constrained and safe RL for wireless

Constrained MDPs [16] and their Lagrangian relaxations underpin most safe-RL practice. PPO [17] is routinely extended with primal-dual updates on constraint slack [18], trust-region feasibility projection [19], or Lyapunov-based stability [20], [21]. In wireless, Khairy *et al.* [22] are the reporting-standard reference for constrained DRL—multiplier-trajectory

plots, 95% confidence bands, and an explicit zero-shot generalization section (train on 200 devices, test on 100/600/1000)—and we adopt their reporting template. A recurring but rarely-audited assumption in this literature is that the environment is feasible, so that the dual variable converges to a bounded shadow price; CMDP theory says exactly the opposite when it is not—under an infeasible constraint the multiplier diverges to its cap [16], [23], and reachability-constrained RL shows that states outside the feasible set need a dedicated feasibility mechanism rather than a penalty [24]. *Boundary*: we operate the *same* dual machinery across a feasible (nominal) and an infeasible (peak-overload) operating condition and report both pathologies empirically—an interior shadow price in the former, a diagnostic ceiling ratchet in the latter—and we contribute the certificate-based escalation/admission layer as the feasibility mechanism that carries the safety load where pricing cannot.

C. U-space separation management

U-space regulation [2] and its concept of operations [25]–[27] define how low-altitude airspace is kept safe. The SESAR JU BUBBLES deliverable [3] formalizes the separation management service: a traffic performance envelope, a chained tactical-contact separation minimum, a target-level-of-safety budget, and numerical scenarios; peer-reviewed near-midair-collision modelling provides the double-cited safety anchors [28]. BUBBLES itself notes that UTM has begun to adopt RL (for learning separation minima), which legitimizes a learning approach in this domain. *Boundary*: prior UTM work treats communication performance as an exogenous requirement (fit-for-purpose, with no hard delay numbers), whereas we close the loop—the semantic-communication choice feeds the communication-delay term of the separation chain, making evidence selection a first-class flight-safety variable. To our knowledge no prior semantic-communication or DRL resource-allocation paper grounds its entire scenario in a U-space deliverable at the parameter level.

III. BUBBLES-CONFORMANT SYSTEM MODEL

We consider N small UAVs operating in a shared low-altitude volume and answering a stream of VQA queries through an edge server over an A2G link. Rather than invent the airspace, mobility, and conflict models, we anchor every parameter to the SESAR JU BUBBLES D2.1 deliverable [3], double-citing peer-reviewed or regulatory sources for load-bearing claims [2], [28].

A. Airspace, envelope, and demand

The operating volume is a 5×5 km area with three flight layers at 20–120 m altitude (BUBBLES Appendix G). The traffic *performance envelope* (Appendix B) fixes the UAV kinematics: cruise speed 5–25 m/s, climb rate 3–5 m/s, and air-frame size 0.5–5 m. The Appendix-G Table G-13 operational demand—daily-mean 8.17 concurrent aircraft, peak 20, 784 operations/day, mean operation duration 15 min—defines two operating conditions we evaluate under:

- **peak (utm_conflict)**: 20 concurrent aircraft, dense arrivals; the conflict constraint is genuinely activated (a stress test of safety behaviour);
- **nominal**: 8.17 concurrent aircraft; the default evaluation condition, where the airspace is uncongested.

Each aircraft is placed on an *Area4D* operational volume [3]—a disk of radius 130 m matching the nominal sensor footprint—on a jittered grid, and its background operational intents are generated from that aircraft’s own task stream through a density gate (activation probabilities 0.30 nominal / 0.12 peak). *Honesty note*: we therefore do *not* inject an independent environmental traffic population; the demand is a *fleet-internal* traffic model in which every intent traces to a served UAV, and a separate exogenous-traffic layer is left to future work. A consequence of the Area4D packing is a scenario-intrinsic baseline geometric-conflict rate—it is a property of the demand geometry, not an artifact of the scheduling policy.

VQA tasks arrive as a VisDrone-derived query stream [29], modelled as an operations-to-queries process in which each active operation emits a question stream over its dwell. Each episode serves 20 queries in one-second decision slots (one service decision per slot); under the peak preset the queries arrive as a front-loaded burst with a 50% critical/high risk mix, while nominal arrivals are staggered.

B. A2G link, bandwidth, and interference

The air-to-ground uplink uses log-distance path loss with an elevation-dependent line-of-sight (LoS) term: we adopt the Al-Hourani LoS sigmoid [30] with its original urban parameters ($a=9.61$, $b=0.16$) and a path-loss exponent of 2.2–2.25, cross-checked against 3GPP TR 36.777 aerial channel models [31]. Shadow fading uses $\sigma_{SF} = 2$ dB (36.777 UMa-AV LoS gives $4.64 e^{-0.0066h} \approx 2.6$ dB at 90 m), and small-scale fading is Rician with $K=6$ dB, within the $K \approx 2$ –15 dB range reported for measured A2G links [32]. The received SNR is quantized to $\{-5, \dots, 20\}$ dB bins.

Bandwidth and multiple access. The A2G bandwidth is not a private per-aircraft channel: it is a *share of a common spectrum pool*, and the continuous resource action allocates a fraction of that pool (0.65–1.0 MHz per served task in our scenario tiers). The main training and evaluation loop is *slotted-serial on the uplink*: each decision slot serves a single task upload (an implementation fact), so concurrent co-channel interference does not accumulate on the main path and the noise floor is the aggregate out-of-system interference power constant I_0 . We sweep I_0 from -120 to -116 dBm as an *interference-floor axis* for zero-shot robustness; this is a raised noise floor, not a claim of resolved multi-user contention. A partially overlapping shared-pool leakage mechanism (concurrent uploads contributing weighted co-channel interference) is exercised only in the multi-UAV concurrency benchmark of Section VI and is left as an extension of the main serial model. Transmission delay uses a capacity-abstraction rate (Shannon spectral efficiency at the operating SNR, with an implementation efficiency gap of $\eta \approx 0.75$ understood but not separately modelled), so quoted uplink times are lower bounds on a fielded modem.

C. Tactical-contact separation chain

Following BUBBLES §3.3.4 (Block 4), the tactical-contact separation minimum is built up from the physical near-mid-air-collision distance by adding error and time-window terms:

$$d_{TC} = d_{NMAC} + TSE + SSE + V_c[(T_1+T_2) + T_{res} + (T_3+T_4)], \quad (1)$$

where d_{NMAC} is the downstream near-collision barrier, TSE/SSE are the total/surveillance system errors, V_c is the closing speed, T_1, T_2, T_3 are detection/decision/manoeuvre time windows, and T_{res} is the separation-recovery time. The final term T_4 is the sum of three normally distributed components—tracking delay, *separation communication delay*, and pilot-execution delay (Table G-2 example: communication mean 1.8 s / σ 1.0, T_4 total mean 7.82 s / σ 4.24). Equation (1) is the mechanistic core of our safety bridge (Section VI): communication delay $\uparrow \Rightarrow T_4 \uparrow \Rightarrow V_c T_4 \uparrow \Rightarrow d_{TC} \uparrow \Rightarrow$ airspace capacity $\downarrow$. Communication delay is not exogenous noise; it is an additive Gaussian component inside the separation minimum.

A pair of UAVs is in *tactical conflict* only when both a spatial and a temporal criterion hold: the closest-point-of-approach (CPA) distance falls below d_{TC} and time-to-CPA is below 60 s. Critically, CPA is extrapolated from the kinematics of the *assigned* UAV (the one tasked to serve), not the nearest arbitrary UAV; using the nearest UAV would make conflicts almost exogenous to the policy, leaving “select cache” as the only avoidance action (see Section V).

D. How payload enters T_4 : an M/G/1 queueing coupling

The separation-communication component of T_4 (Table G-2 mean 1.8 s) is the tactical loop’s own exchange—formulate the resolution, deliver the instruction. Our VQA payload does not *replace* that exchange; it *contends* with it on the shared RAN. We therefore treat the 1.8 s as the queue-empty tactical baseline and add the contention as the mean waiting time W of a non-preemptive priority M/G/1 queue [33]: the separation-relevant (C2-class) message is the high-priority class, payload uploads are the low-priority class whose service time S is the airtime of the evidence level the scheduler chose, and per-class utilisations follow the G-13 demand ($\rho_{C2}/\rho_{payload} = 0.10/0.35$ nominal, 0.25/0.55 peak). Then

$$T_4^{\text{comm}} = 1.8 \text{ s} + W, \quad W = W_{M/G/1}(S(\text{evidence}), \rho), \quad (2)$$

so a heavy upload lengthens the *residual service* a tactical message must wait behind, and the evidence choice reaches d_{TC} through the queue rather than by substitution. Because whether C2 shares the payload link is contested (Section III-E), we evaluate both a *shared* mode (C2 in the pool as the top class) and a *dedicated* mode (C2 on its own band; only same-class payload contention remains). Table I shows W at the -5 dB worst case: a token upload (14 ms airtime) adds 3–8 ms of wait—the tactical loop does not notice—while an image upload (3.72 s airtime) adds 0.78–1.98 s, a 44–110% inflation of the separation-communication mean.

TABLE I: M/G/1 wait W (s) added to the 1.8 s separation-comm baseline at -5 dB Rayleigh, by load and C2 spectrum mode.

Evidence	shared C2		dedicated C2	
	nominal	peak	nominal	peak
s_1 token ($S=0.014$ s)	0.004	0.008	0.003	0.005
s_2 image ($S=3.72$ s)	0.930	1.984	0.783	1.435

E. Deadline QoS and the C2 priority class

Each query carries a delay budget τ_k interpreted as the *communication-and-decision window*: the time from question arrival to a returned answer usable by the separation service. Flight-repositioning time is a *tasking-layer* quantity and is deliberately *not* charged to τ_k . BUBBLES gives no hard communication-latency numbers—a fit-for-purpose principle citing ED-261 GEN-SUR (D2.1 p.62)—so rather than assert a deadline we anchor it to the separator-communication delay of the tactical chain: Table G-2 models that delay as $\mathcal{N}(1.8, \sigma 1.0)$ s, and we set the critical and normal budgets at its $+2\sigma/+3\sigma$ confidence points, $\tau_{\text{critical}} = 3.8$ s and $\tau_{\text{normal}} = 4.8$ s. Treating the separation-relevant semantic evidence stream as the highest-priority class within the task-data-link QoS is consistent with the C2 key-performance-indicator targets of 3GPP TS 22.125 [34] and with EUROCAE ED-282 as supporting guidance [35]. We stress the scope red line: authenticated command-and-control / control-and-non-payload-communication (CNPC) runs on protected aeronautical spectrum (ICAO WRC-12 5030–5091 MHz) and is *outside* this paper; a structural precedent for isolating such a flow as a separate QoS flow on the same RAN is 3GPP TS 22.125 / TS 23.256 [34], [36].

F. Spec-attainability certificate and the escalation budget

Peak overload creates queries that *no* action can serve within spec: at a weak link even the best transmission service cannot clear the quality threshold inside the deadline. Charging such physically unserveable queries to the policy as quality violations would poison both the learning signal and the safety accounting, so we make unserveability an explicit, certified state.

Definition 1 (Spec-attainability certificate). *At slot t , query k with remaining window $\tilde{\tau}_k(t)$ and quality threshold ε_k is spec-attainable, $A_k(t) = 1$, iff some transmission service $s \in \{s_1, s_2\}$ satisfies both $\text{LCB}(q_s(\gamma_k)) \geq \varepsilon_k$ and $D_s^{\text{comm}}(\gamma_k) \leq \tilde{\tau}_k(t)$, where q_s is the calibrated quality of service s at link state γ_k , LCB its lower confidence bound, and D_s^{comm} the communication-window delay of Section III-E.*

The certificate is an *environment* property (calibrated quality tables plus link physics, independent of the learner), is exposed to the policy as a state feature, and is recomputed *every* slot from the remaining window—a query whose window has been eaten by deferrals turns unattainable mid-episode. It drives a certified *escalation* path for the safety-critical class: when a critical/high query ends the episode rejected or expired, the certificate decides the verdict. If $A_k = 1$ the policy dropped a

serveable safety query and is charged the full quality violation; if $A_k = 0$ the query is *escalated*—handed to a higher-tier service (relay tasking, human review) outside this scheduler—carrying no quality cost but consuming a budgeted escalation event. Escalation is priced by a dedicated dual channel with budget δ_{esc} (Section IV).

The budget is *derived, not tuned*: rolling the certificate over the scenario arrival mix measures the structurally unserveable fraction of the critical load (0.105 under peak, 0.000 under nominal), and δ_{esc} adds a 0.05 headroom: $\delta_{\text{esc}}^{\text{peak}} = 0.155$, $\delta_{\text{esc}}^{\text{nominal}} = 0.05$. Queries that are not escalated form the *admitted set*; all rate constraints in Section IV are stated on it. This is the standard demanded-versus-achievable split of admission control: the scheduler is accountable for what it admits, and the certificate makes the un-admittable load visible, budgeted, and auditable rather than silently absorbed into violation statistics.

G. TLS anchoring and the conflict budget

The target level of safety is a total $\text{TLS} = 10^{-6}$ fatal accidents per flight hour (FAT/FH), of which the mid-air-collision share is $\text{TLS}_{\text{MAC}} = 2.5 \times 10^{-7}$ FAT/FH; collision risk scales as $N(N-1)$ under the gas-kinetic model, and a severity ladder $\text{MAC} \leftarrow \text{NMAC} \leftarrow \text{IC} \leftarrow \text{SL} \leftarrow \text{TC}$ places one mitigation barrier at each level. We use this ladder to *derive* a per-slot conflict budget rather than assert one: a tactical-contact violation sits at the top of the ladder, protected by several downstream barriers, so it may occur far more frequently than a MAC. Normalizing that admissible frequency by the peak concurrency and the slot window yields a per-slot tactical-conflict budget of ≈ 0.08 , which we use as the conflict constraint limit in Section IV. *Honesty note*: BUBBLES gives no hard communication reliability/latency numbers (a fit-for-purpose principle), so the specific normalization constant behind 0.08 is our engineering design choice within the barrier-ladder+demand framework, declared as such rather than read directly from the deliverable.

H. Separation kernel and validation gate

The chain (1), the TLS anchoring, and the CPA dual criterion are implemented in a standalone separation kernel gated by a reproduction test against the official numerics: it recomputes BUBBLES Tables G-4/G-5 cell-by-cell to within 0.18% (e.g., a total contact distance of $370.53 \rightarrow 370.49$ m for a SAIL I-II scenario), and a smoke scenario injecting an approaching pair lifts the measured conflict rate from 0 to 0.5, confirming the dual criterion fires; the full suite passes 14/14 unit tests. The kernel is a fixed “consumer” that receives the measured link delay from the semantic layer and returns separation distances and conflict flags.

I. Semantic evidence levels

The transmitter chooses, per query, one of three evidence levels (a fourth, region-of-interest crop, is retained as an optional action but excluded from the main comparison); the receiver is a frozen commercial vision-language model

(Qwen2-VL-2B) [37]. This taxonomy and its per-question complementarity are established in the companion paper [1]; we summarize the transmission cost, since it drives both the resource and safety constraints.

- s_0 **cache**: reuse a fresh cached answer; 0B, no transmission, and—in the BUBBLES setting—*zero conflict exposure*, since no flight/communication action is initiated. A cache entry ages fresh→stale→expired and cannot be reused past expiry; its effective accuracy is $A_{s_0} = A_{\text{LUT}}(0.85 + 0.15 p_{\text{hit}})$.
- s_1 **token**: light detector tokens (class/box/confidence), ~ 0.7 – 1.2 KB, framed under LDPC rate-1/2 with BPSK [38], costing $N_{s_1} = 16 B_{\text{tok}}$ complex channel uses ($\sim 11\text{k}$ – 19k), decoded by a symbolic reader or the VLM.
- s_2 **image**: full JPEG at rate-adaptive quality, ~ 110 KB, costing $N_{s_2} = 8 B_{\text{JPEG}} / \text{SE}(\bar{\gamma})$ uses; at -5 dB this reaches 3.72M uses ($\approx 265\times$ the token cost) and a 3.72 s uplink, answered by the VLM.

The two-order-of-magnitude gap between s_1 and s_2 under weak links is what makes evidence selection a scheduling lever with both a resource-budget and a flight-safety effect.

IV. PROBLEM FORMULATION

We model joint semantic-evidence and airspace-resource scheduling as a constrained Markov decision process (CMDP) $\langle \mathcal{S}, \mathcal{A}, P, r, \{c_i\}, \{d_i\}, \gamma \rangle$ with a *budgeted escalation channel*. The certificate of Section III-F splits the query stream into the admitted set $\mathcal{K}_{\text{adm}}(\pi)$ (served, rejected-while-serveable, or expired-while-serveable) and the escalated remainder; the rate constraints are stated *on the admitted set*, and escalation itself is a budgeted event:

$$\begin{aligned} \max_{\pi} \mathbb{E} \left[\sum_t \gamma^t r(s_t, a_t) \right] \\ \text{s.t. } \mathbb{E}[c_i \mid k \in \mathcal{K}_{\text{adm}}(\pi)] \leq d_i, \quad i \in \mathcal{I}, \\ \mathbb{E}[\mathbf{1}\{k \text{ escalated}\}] \leq \delta_{\text{esc}}, \end{aligned} \quad (3)$$

with $\mathcal{I} = \{\text{quality-n, quality-c, deadline-n, deadline-c, conflict, battery, GPU}\}$. This admitted-set formulation is the honest one for an overloaded safety system: a scheduler cannot promise a violation rate on queries that are physically unserveable, but it can promise (i) a violation rate on what it admits and (ii) a certified, budgeted escalation rate for what it does not. Both promises are auditable, and Section VI reports them channel-by-channel.

A. State

The recommended observation `state_v2_fixed` is ≈ 137 -dimensional and comprises: (i) a task block (question-type one-hot, SNR, deadline, cache-hit rate, and the spec-attainability certificate bit of Definition 1); (ii) per evidence-level features (LUT payload and accuracy, plus seven semantic-feasibility features per level—accuracy lower-confidence bound, uncertainty, quality gap, joint feasibility); (iii) per-UAV state (7 dims: position, battery, status) and per-edge state (4 dims: CPU, GPU, VRAM, queue); (iv) the water levels of five Lyapunov virtual queues; and (v) mobility

TABLE II: Hybrid action space and control timescales.

Head	Type	Controls / timescale
service	disc.	evidence route (cache / token / image);
	$\{0, 1, 2\}$	fast (each slot)
resource	cont.	A2G bandwidth share, Tx power
	4-d	$[0.05, 1]$ W, edge CPU, edge GPU
		share; fast
mobility	disc.	task-UAV assignment; 5 manoeuvre
		modes (stay, serve, reposition, avoid,
		return-base); slow ($K=3$)
waypoint	cont.	$\tanh \times 80$ m horizontal, $\tanh \times 20$ m
	2-d	vertical increment; slow

features and action-mask features stored for replay at update time (Section V).

B. Hybrid action space (two timescales)

The action factorizes into a fast head, updated every slot, and a slow head, updated every $K = 3$ slots (Table II). The fast head carries the semantic-evidence routing and physical-resource shares; the slow head carries task assignment and flight manoeuvres, whose effects (delay, energy, conflict geometry) unfold over multiple slots. Continuous sub-actions pass through a state-independent Gaussian, then a sigmoid/tanh squash into their physical ranges. Two further outcomes complete the action semantics: *reject/defer*, produced when the semantic feasibility projection (Section V) rewrites a deadline-infeasible transmission into an explicit drop or postponement (a first-class action with its own reward accounting), and *escalate*, which is *not* policy-selectable—the certificate of Definition 1 adjudicates every critical/high query that ends rejected or expired, so the policy cannot game the escalation channel by dropping serveable safety queries.

C. Constraint set

The constraints are risk-differentiated: *critical* tasks carry tighter limits than *normal* ones (Table III). The quality constraints cap the semantic-quality violation rate on the admitted set; the deadline constraints cap the timeout-violation rate; the conflict constraint caps the tactical conflict rate at the TLS-derived budget of Section III; battery and GPU constraints cap resource-exhaustion risk; and the escalation budget caps how much of the critical load may be handed upward.

D. Reward and environment

The reward is a heavily shaped `semantic_utility` signal: semantic success and an accuracy-lower-confidence-bound margin (risk-weighted, with critical tasks scaled $\times 1.6$), minus a quality-gap penalty, delay/energy/payload costs, and manoeuvre/safety costs; a cache-collapse penalty ($\sim -16 \times \text{gap}$ when s_0 is selected despite a semantic gap) discourages degenerate reuse. The environment models a six-term delay decomposition (fly+sense+tx+queue+infer+load), a four-term energy model, GPU VRAM with an LRU model cache, the fresh→stale→expired semantic cache, and the UTM/DSS conflict state machine; the A2G propagation, shared-pool bandwidth share, and slotted-serial uplink follow Section III-B,

TABLE III: Constraint set and limits d_i (rates on the admitted set; escalation is a budgeted channel of its own).

Constraint c_i	Limit d_i	Source
quality (normal)	0.05	risk-aware setting
quality (critical)	0.02	risk-aware (tighter)
deadline (normal)	0.05	$\tau_{\text{normal}}=4.8$ s QoS (§III-E)
deadline (critical)	0.02	$\tau_{\text{critical}}=3.8$ s QoS (§III-E)
conflict	≈ 0.08	TLS _{MAC} via barrier ladder (§III)
battery	—	UAV energy model
GPU (VRAM)	—	model-cache LRU
escalation	0.155 / 0.05	certificate calibration, peak/nominal (§III-F)

with received SNR quantized to $\{-5, \dots, 20\}$ dB bins. This CMDP is the target of the constrained-RL solver in Section V.

V. RISK-AWARE CONSTRAINED HYBRID-ACTION PPO WITH DUAL-PRICE MONITORING

We name the method precisely: a *risk-aware constrained hybrid-action PPO with a certified escalation layer and dual-price diagnostics*, trained on the two-timescale action space of Section IV with a slow mobility-semantic actor. The name reflects the division of labour established in Section VI: in the feasible operating regime the Lagrangian duals converge to interior shadow prices, and in the infeasible overload regime the certificate-based escalation/admission layer carries the safety guarantee while the dual's ceiling ratchet serves as an infeasibility diagnostic. This section covers the network and training (§V-A), the six-layer constraint machinery (§V-B), the correctness fixes and the ratio $\equiv 1$ test gate (§V-C).

A. Network and training

A shared MLP encoder (128, 128; ReLU) feeds single-layer policy heads and a value head; continuous heads use a state-independent learnable $\log \sigma$. We optimize with Adam ($\text{lr } 3 \times 10^{-4}$), PPO clip 0.2 [17], and four epochs per episode. Advantages use Monte-Carlo discounted returns minus the shared value baseline (no GAE): episodes are short (20 steps) and rewards dense, so GAE's bias-variance trade buys little and costs a hyperparameter; the price is higher variance, noted as a limitation. Entropy is annealed linearly $0.10 \rightarrow 0.02$ to suppress premature collapse to the cache action, and a behaviour-cloning (BC) warm start from greedy demonstrations is applied through a decaying auxiliary loss. Rollout collection is wrapped in `no_grad`. Training configuration is summarized in Table IV.

B. Six-layer constraint machinery

Constraint handling is realized by six stacked layers. The division of labour, established empirically in Section VI, is that the *structural* layers (2–5) plus the *admission* layer (6) carry the safety load, while the *dual* layer (1) prices the constraints where they are feasible and diagnoses them where they are not.

TABLE IV: Training configuration.

Component	Setting
Shared encoder	MLP (128, 128) + ReLU; policy/value share encoder
Action heads	single linear layer each; state-indep. log σ
Optimizer	Adam, lr 3×10^{-4}
PPO clip / epochs	0.2 / 4 per episode
Advantage	MC return $- V$ baseline; scale-only norm. (no GAE)
Entropy coeff.	0.10 $\rightarrow$ 0.02 linear anneal (over 120 ep)
BC warm start	greedy demos, 50 ep; bc_aux 0.05 + service prior 0.65, both time-decayed
Dual schedule	lr 0.1 (0.2 conflict); caps $\lambda^{\max}=20$, per-channel 8 (quality, conflict); leak 0.01; warm-up 150 ep
Episode	20 decision slots (1 s each)
Protocol	3 seeds $\times$ 500 train ep; test 50 rollouts/point

- 1) **Lagrangian dual** (eight multipliers λ_i , incl. the escalation channel): projected dual ascent $\lambda_i \leftarrow [(1-\kappa)\lambda_i + \eta_i(c_i - d_i)]_0^{\lambda_i^{\max}}$ with risk-aware limits, ascent rates $\eta_i = 0.1$ (0.2 for the conflict channel), a leak $\kappa=0.01$ (releasing the ratchet so λ can decrease; disabled on the quality/escalation channels, which are pure projected sub-gradients), per-channel ceilings ($\lambda_Q^{\max}=\lambda_{\text{conf}}^{\max}=8$ vs. the global 20), and a 150-episode dual warm-up during which ascent is frozen so the policy forms under the BC/prior shaping before the constraint prices start to move.
- 2) **Action mask**: hard feasibility masks (GPU-VRAM masks the service level, battery masks the UAV, and manoeuvre modes are masked by state).
- 3) **Semantic feasibility projection**: on a hard violation the service is re-selected by a semantic safety score; a cache-override forces an evidence upgrade under a large semantic gap, and a deadline-infeasible transmission is rewritten into token $\rightarrow$ cache $\rightarrow$ reject $\rightarrow$ defer, the reject/defer outcomes of Section IV. A structural *critical-cache ban* rules a critical/high query served from cache alone never quality-compliant, closing the degenerate cache shortcut.
- 4) **Resource-floor projection**: per-route lower bounds on bandwidth/power/CPU/GPU; resources are zeroed for s_0 .
- 5) **Lyapunov virtual queues** ($\times 5$): quality, deadline, energy, risk, and UTM queues enter the observation and a drift-plus-penalty term [20], [21]; updated incrementally.
- 6) **Certified escalation/admission** (Section III-F): the spec-attainability certificate adjudicates every critical/high query that ends rejected or expired—attainable ones are charged as quality violations, unattainable ones are escalated against the budget δ_{esc} through the dedicated dual channel. This is the layer that *creates* a feasible region in the overload regime (Section VI).

C. Correctness audit and the ratio $\equiv 1$ gate

A two-model independent read-only audit (cross-validated, zero contradictions) surfaced defects that had contaminated an

Algorithm 1 Risk-aware constrained hybrid-action PPO with certified escalation

```

1: Init. shared encoder, heads, value;  $\lambda_i \leftarrow 0$ ; BC warm start.
2: for episode = 1, ..., 500 do
3:   Collect rollout (20 slots) under no_grad; every slot run fast head, every  $K=3$  slots run slow head; apply masks (L2), feasibility projection incl. reject/defer rewrite (L3), resource floors (L4).
4:   Certificate adjudication (L6): critical rejected/expired queries are charged (attainable) or escalated against  $\delta_{\text{esc}}$  (unattainable).
5:   Update Lyapunov queues (L5); log costs  $c_i$  and multipliers  $\lambda_i$ .
6:   for epoch = 1, ..., 4 do
7:     Recompute log-probs; assert epoch-0 ratio $\equiv 1$  (unit gate).
8:     PPO clip loss + Lagrangian term  $\sum_i \lambda_i c_i$  + Lyapunov drift + decayed BC aux.
9:     Scale-only advantage normalization (preserve mean; keep  $\lambda$  penalty).
10:   end for
11:   if episode > 150 (dual warm-up over) then
12:      $\lambda_i \leftarrow \text{clip}((1-\kappa)\lambda_i + \eta_i(c_i - d_i), 0, \lambda_i^{\max})$  per channel (L1).
13:   end if
14: end for

```

earlier “two-timescale vs. monolithic” ablation. We enumerate the load-bearing findings and their fixes.

- **Slow-head credit assignment (P0)**: the slow mobility head’s log-prob was counted every step ($\times K$), and its action mask was applied at sampling but not at update, so the policy ratio $\neq 1$ even at epoch 0, spuriously triggering the PPO clip and mis-counting entropy. *Fix*: count the log-prob once per decision step (zero on non-decision steps), store the mobility mask for replay, and multiply entropy by the decision mask.
- **Two-timescale mobility double-count (P0)**: the mobility cost was already inside the semantic_utility reward, and the two-timescale path added it again, so the old structural ablation partly measured a reward-definition difference. *Fix*: remove the duplicate adjustment and rerun.
- **Multiplier ratchet (P1)**: with a zero limit and non-negative cost, λ was monotone non-decreasing—a barrier, not a shadow price. *Fix*: add the leak term and per-channel caps.
- **Penalty double-counting (P1)**: a conflict set both an airspace and a UTM flag, stacking a shaped penalty with the Lagrangian and Lyapunov terms. *Fix*: penalize the same event once; zero the UTM channel weight; make queues incremental.
- **Assigned-UAV CPA (P1-env)**: CPA used the nearest UAV, making conflicts nearly exogenous. *Fix*: use the assigned UAV’s kinematics, restoring policy controllability of conflict.
- **BC lock-in / advantage normalization / efficiency (P1-P2)**: decay the BC auxiliary weight; use scale-only advantage normalization so the near-constant λ penalty is not subtracted away; wrap rollout collection in no_grad.
- **Reward-attribution wiring (P0, found post-audit)**: the mission-aligned success attribution (a quality- and deadline-compliant service that is merely UTM-blocked earns a discounted success bonus instead of a penalty) was applied in the environment log but *not* in the learner’s shaped reward, so the gradient never saw it. *Fix*: mirror

the attribution into the controller reward; the unconstrained arm’s peak behaviour then improves measurably (Section VI-D), which doubles as an end-to-end witness that the reward path is load-bearing.

The standing regression gate is an **epoch-0 ratio $\equiv$ 1 unit test**: immediately after collection we recompute the log-probs and assert they equal the stored old log-probs step-by-step, guaranteeing the PPO importance ratio starts at exactly 1. The full suite passes 321 tests, including bit-for-bit legacy-regression guards on every constraint-machinery lever (escalation, cache ban, dual caps, warm-up, reward attribution). Per the audit, the results in Section VI are all on the corrected system.

VI. EXPERIMENTS

We follow the reporting template of constrained-DRL practice [12], [22]: a convergence triptych with the dual trajectories, a dual-condition main comparison with oracle upper bounds and non-learning baselines, per-channel constraint accounting, a dual-machinery ablation, four-axis generalization sweeps with a dedicated zero-shot section, and the corrected safety bridge.

A. Protocol and metrics

All runs use the frozen environment of Sections III–V (communication-window deadlines, attainability-calibrated ε_k , certified escalation, structural cache ban, mission-aligned reward attribution, fair-share reference bandwidth, LUT out-of-support outage guard, per-channel dual caps with warm-up). Learning arms train 500 episodes $\times$ 3 seeds and are evaluated deterministically on 50 held-out episodes (1,000 queries) per condition; non-learning baselines use the same evaluation. The peak-condition escalation budget is $\delta_{\text{esc}} = 0.155$ and the nominal budget 0.05 (Section III-F). We report: **mission** (quality- and deadline-compliant fraction of the *admitted* set, the honest headline of Eq. (3)), **task** (end-to-end task success), **acc** (mean answer accuracy on the admitted set), **cacheR** (cache-route share), **neRejR** (non-escalated rejects: serveable queries the policy dropped), **critEsc** (critical-escalation rate vs. its budget), **ddlVio** (admitted-set deadline violations), and **conflict** (tactical-conflict rate vs. the 0.08 TLS budget).

Two oracles bound the problem from above. The *best-feasible oracle* picks, per query, the cheapest evidence level that succeeds end-to-end, but never escalates; the *escalation-aware oracle* additionally honours the certificate—a critical query with no spec-attainable transmission service is escalated rather than served badly. The gap between them measures what admission structure is worth.

B. Main dual-condition comparison

Table V reports the main comparison.

Peak is an admission regime. The escalation-aware oracle reaches admitted mission 0.916 (admitted deadline violations 0.084)—but only by escalating 0.666 of the critical load. The best-feasible oracle, denied escalation, collapses to 0.528 with admitted deadline violations of 0.412: *even an oracle cannot keep the mission constraint on the admitted set without shedding load*. The feasible region for a non-escalating strategy is

essentially empty at peak, and safety in this regime is created by the certificate-plus-escalation structure, not by any strategy-layer optimization. Among learning arms, the constrained and fixed-penalty policies land at admitted mission 0.180/0.184 with near-zero conflict (0.003, $27\times$ under budget) and tight admitted deadline compliance (0.023), but they over-reject ($\text{neRejR} \approx 0.35$)—dropping serveable queries is the cheap escape from a pinned quality price (Section VI-D)—while the unconstrained arm escalates $2\text{--}4\times$ more (0.329, over budget) and rejects less. The flat single-head baseline is instructive in the opposite direction: it trains stably (0.344 admitted mission) but *never interacts with the admission machinery at all* (zero rejects, zero escalations)—it serves structurally unserveable critical queries rather than handing them up, its admitted quality-violation rate is 0.564, and its conflict rate is $22\times$ the proposed policy’s (0.065 vs. 0.003, still under budget). No learning arm approaches the escalation-aware ceiling; we return to what peak *does* certify in Section VI-C.

Nominal is a pricing regime. Under the feasible nominal condition the proposed policy reaches admitted mission 0.653—93% of the escalation-aware oracle’s 0.703 and above semantic-greedy’s 0.626—with zero escalations and zero conflicts; trained directly on nominal it reaches 0.723, and the fixed-penalty arm (its static quality price clamped to the same per-channel cap) reaches 0.727. The flat baseline trails the hybrid arms (0.592), and the unconstrained arm gives up deadline discipline (admitted deadline violations 0.254 vs. the proposed 0.168 and nominal-trained 0.047). The backend ablation (legacy LUT quality instead of the calibrated v5 table) shifts absolute numbers (0.443 nominal) without changing any ordering, so claims are conditional on the calibration but not artifacts of it.

C. Per-channel constraint accounting

Table VI reports every dual channel’s realized evaluation cost against its limit, in the implemented constraint semantics (joint risk-class event rate on the admitted set). The pattern is the paper’s core empirical claim. *Structural channels hold everywhere*: conflict (0.003 peak, 0.000 nominal, vs. 0.08), battery, GPU, both critical-deadline channels at peak, and the escalation budget (0.086 vs. 0.155 peak; 0 vs. 0.05 nominal). *Quality channels expose the regime split*: at peak the realized quality cost sits an order of magnitude above its limit (0.41 vs. 0.02 critical)—the infeasible domain, where the accounting shows exactly *which* promise cannot be made and the certificate/escalation layer bounds what happens to the unserveable load instead. Under nominal the quality costs drop to 0.08–0.12: still above the tight risk-aware limits at episode 500, with an interior, still-climbing dual pricing the residual gap (Section VI-D)—approaching, not attained, and stated as such. We emphasize what this table replaces: an earlier draft of this work claimed “constraint satisfaction by construction”; the honest, channel-resolved statement is that the *hard-structure* channels are satisfied by construction, the escalation budget is satisfied by certificate, and the quality channels are priced but not closed.

TABLE V: Main dual-condition comparison on the frozen v8 environment (mean $\pm$ std over 3 seeds where applicable; oracles and heuristics are deterministic). Mission/acc/ddlVio are on the admitted set (Eq. (3)); critEsc is against the escalation budget (0.155 peak / 0.05 nominal); conflict is against the 0.08 TLS budget.

Cond.	Method	Mission (adm.)	Task	Acc (adm.)	CacheR	neRejR	CritEsc	DdlVio (adm.)	Conflict
peak	Esc-aware oracle	0.916	0.789	0.808	0.000	0.000	0.666	0.084	0.193
	Best-feasible oracle	0.528	0.260	0.757	0.008	0.000	0.000	0.412	0.612
	Semantic greedy	0.616	0.268	0.719	0.008	0.000	0.000	0.380	0.612
	Always cache	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000
	Random	0.292 $\pm$ 0.007	0.120 $\pm$ 0.017	0.454 $\pm$ 0.010	0.339 $\pm$ 0.019	0.000	0.000	0.333 $\pm$ 0.022	0.419 $\pm$ 0.027
	Flat PPO (single head)	0.344 $\pm$ 0.051	0.295 $\pm$ 0.052	0.361 $\pm$ 0.021	0.549 $\pm$ 0.026	0.000	0.000	0.093 $\pm$ 0.036	0.065 $\pm$ 0.023
	No-Lagrangian	0.164 $\pm$ 0.015	0.384 $\pm$ 0.226	0.213 $\pm$ 0.074	0.298 $\pm$ 0.193	0.287 $\pm$ 0.076	0.329 $\pm$ 0.271	0.093 $\pm$ 0.097	0.005 $\pm$ 0.002
	Fixed-penalty	0.184 $\pm$ 0.013	0.214 $\pm$ 0.024	0.168 $\pm$ 0.007	0.452 $\pm$ 0.029	0.327 $\pm$ 0.040	0.081 $\pm$ 0.016	0.021 $\pm$ 0.009	0.003 $\pm$ 0.002
	Proposed	0.180 $\pm$ 0.011	0.210 $\pm$ 0.023	0.166 $\pm$ 0.005	0.439 $\pm$ 0.035	0.345 $\pm$ 0.042	0.086 $\pm$ 0.021	0.023 $\pm$ 0.010	0.003 $\pm$ 0.002
nominal	Esc-aware oracle	0.703	0.703	0.454	0.000	0.033	0.000	0.234	0.000
	Best-feasible oracle	0.691	0.691	0.485	0.037	0.000	0.000	0.281	0.000
	Semantic greedy	0.626	0.626	0.461	0.052	0.000	0.000	0.339	0.000
	Always cache	0.250	0.250	0.000	0.750	0.000	0.000	0.000	0.000
	Flat PPO (single head)	0.592 $\pm$ 0.074	0.592 $\pm$ 0.074	0.441 $\pm$ 0.018	0.162 $\pm$ 0.054	0.000	0.000	0.263 $\pm$ 0.112	0.000
	No-Lagrangian	0.593 $\pm$ 0.072	0.593 $\pm$ 0.072	0.438 $\pm$ 0.048	0.151 $\pm$ 0.083	0.000	0.000	0.254 $\pm$ 0.148	0.000
	Fixed-penalty	0.727 $\pm$ 0.017	0.727 $\pm$ 0.017	0.409 $\pm$ 0.011	0.155 $\pm$ 0.017	0.000	0.000	0.049 $\pm$ 0.031	0.000
	Proposed	0.653 $\pm$ 0.097	0.653 $\pm$ 0.097	0.436 $\pm$ 0.022	0.153 $\pm$ 0.067	0.000	0.000	0.168 $\pm$ 0.159	0.000
	Proposed (nominal-trained)	0.723 $\pm$ 0.017	0.723 $\pm$ 0.017	0.421 $\pm$ 0.010	0.170 $\pm$ 0.060	0.000	0.000	0.047 $\pm$ 0.028	0.000

TABLE VI: Per-channel realized eval cost vs. limit d_i (admitted set; $\checkmark$ within limit, $\times$ above). Peak/nominal columns: the peak-trained policy in its own condition and evaluated on nominal; nom-tr = trained on nominal.

Channel c_i	d_i	peak	nominal	nom-tr
quality (normal)	0.05	0.389 $\times$	0.101 $\times$	0.109 $\times$
quality (critical)	0.02	0.408 $\times$	0.083 $\times$	0.121 $\times$
deadline (normal)	0.05	0.022 $\checkmark$	0.090 $\times$	0.001 $\checkmark$
deadline (critical)	0.02	0.001 $\checkmark$	0.077 $\times$	0.046 $\times$
conflict	0.08	0.003 $\checkmark$	0.000 $\checkmark$	0.000 $\checkmark$
battery	—	0.000	0.000	0.000
GPU (VRAM)	—	0.000	0.000	0.000
escalation	0.155/0.05	0.086 $\checkmark$	0.000 $\checkmark$	0.000 $\checkmark$

D. Convergence and the dual-pathology contrast

Fig. 1 shows the convergence triptych—episode return, the quality-critical constraint cost against its 0.02 limit, and the dual trajectories—for the peak-trained and nominal-trained proposed policy (3 seeds, $\pm$ std). The dual warm-up (episodes 0–149, shaded) freezes ascent while the BC/prior shaping decays; what happens after it lifts is the paper’s central diagnostic, isolated in Fig. 2. Under *peak*, λ_{QC} resumes from ≈ 0 and ratchets monotonically—1.95 by episode 200, 5.48 by 300, pinned at the per-channel cap 8.00 by episode ≈ 370 on 3/3 seeds with zero fall-back—against a realized cost stuck at ≈ 0.4 . Under *nominal*, the same machinery climbs $\approx 4\times$ slower and stays interior: 0.50 at episode 200, 1.33 at 300, 3.29 at 500, never touching the cap, against a cost declining toward its limit (0.053 at the training tail). One algorithm, one schedule, two textbook behaviours: an interior shadow price where the constraint is feasible, and the classical divergence of the multiplier where it is not [16], [23]. The pin is robust across three engineering generations (caps 20/20/8 pin at 18.3/18.4/8.0) and across the quality backend (2/2 seeds under the legacy LUT), while the constrained arm’s peak behaviour is invariant under all of it (admitted mission 0.179/0.179/0.180)—the levers move the ceiling, not the feasibility.

E. Dual-machinery ablation

Table VII ablates the dual machinery itself at peak (3 seeds each). **(i) Quality channel off** ($\lambda_Q^{\max}=0$): removing the quality price entirely leaves the admitted-set mission statistically unchanged (0.155 vs. 0.180)—in the infeasible domain the price was not buying compliance—but the behavioural split moves toward the unconstrained arm (cache 0.304, escalation 0.230 and over budget): the pinned price had been shaping *how* the policy sheds load, not *whether* it satisfies quality. **(ii) Warm-start at the pin** ($\lambda_{QC}(0)=8$, no warm-up): starting the dual where it will end leaves the headline unchanged (admitted mission 0.177 vs. 0.180; λ_{QC} enters at 8.00 and never leaves on 3/3 seeds)—the pin is the domain’s destiny, not an artifact of the ascent schedule. The one visible effect is again on the load-shedding split: with the full price active from episode 0 the critical-escalation rate lands at 0.134, *inside* the certificate band [0.105, 0.205], where the warmed-up reference under-escalates (0.086)—corroborating that in an infeasible domain the dual’s schedule shapes *how* load is shed, not *whether* the constraint is met. **(iii) No slow head**: removing the learned mobility actor (deterministic mobility defaults) also leaves the headline unchanged (0.176) and even zeroes the residual conflict rate (0.000 vs. 0.003; both $\ll 0.08$)—in this admission-dominated regime the mobility hierarchy is *not* what suppresses conflict, and we retract the stronger causal reading an earlier draft attached to it. Its visible effect is, once more, on the shedding split (critical escalation 0.168, the top of the band). The three ablations tell one story: at peak, *no* strategy-layer knob moves the admitted-mission headline (all arms sit in 0.155–0.184); each knob only redistributes how the unserveable load is shed among escalate/reject/cache. The controlling structure is the certificate and its budget—which is exactly what the admission-regime reading predicts.

F. Sample efficiency

Table VIII reports episodes to 90% of the terminal smoothed return (min-to-terminal span) on a uniform 500-episode budget. The BC warm start front-loads the peak arms (the return

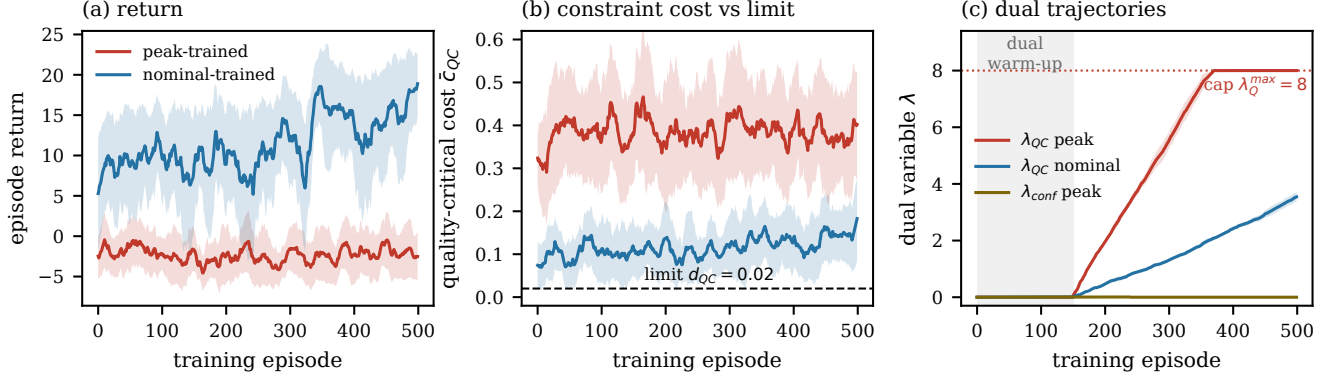


Fig. 1: Convergence triptych on the frozen v8 environment: (a) episode return, (b) quality-critical constraint cost vs. its 0.02 limit, (c) dual trajectories (λ_{QC} , and $\lambda_{conflict}$ at peak), peak-trained vs. nominal-trained, 3 seeds $\pm$ std; dual warm-up window shaded.

TABLE VII: Dual-machinery ablation at peak (mean $\pm$ std over 3 seeds). λ_{QC}^{end} lists the terminal quality-critical dual per seed; the reference row is the proposed v8 arm.

Arm	Mission (adm.)	Task	CacheR	neRejR	CritEsc	DdlVio (adm.)	Conflict	λ_{QC}^{end}
Proposed (reference)	0.180 ± 0.011	0.210 ± 0.023	0.439 ± 0.035	0.345 ± 0.042	0.086 ± 0.021	0.023 ± 0.010	0.003 ± 0.002	8.00/8.00/8.00
(i) quality channel off	0.155 ± 0.025	0.276 ± 0.115	0.304 ± 0.144	0.392 ± 0.044	0.230 ± 0.213	0.051 ± 0.036	0.004 ± 0.004	0.00/0.00/0.00
(ii) warm-start at the pin	0.177 ± 0.016	0.229 ± 0.004	0.444 ± 0.012	0.320 ± 0.015	0.134 ± 0.037	0.021 ± 0.006	0.003 ± 0.002	8.00/8.00/8.00
(iii) no slow mobility head	0.176 ± 0.017	0.246 ± 0.015	0.435 ± 0.025	0.316 ± 0.018	0.168 ± 0.057	0.018 ± 0.002	0.000	8.00/8.00/8.00

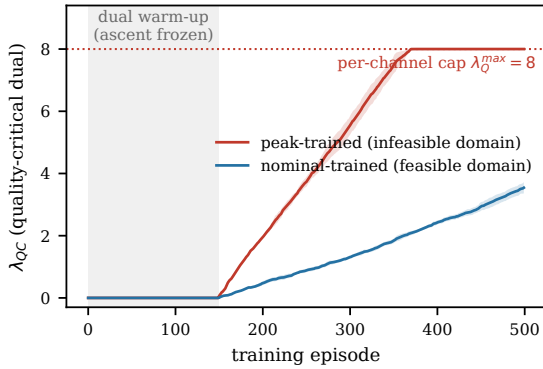


Fig. 2: The dual-pathology contrast: λ_{QC} ratchets to its cap under peak (infeasibility signature) and converges interior under nominal (shadow price), same machinery and schedule.

span is nearly flat from episode 0), so their ep-to-90% is small by construction; the nominal-trained arm, whose return genuinely improves over training, reaches 90% at 146 ± 123 episodes, and the flat baseline at 27 ± 18 . Wall-clock is ≈ 3 minutes per 500-episode arm on a single consumer GPU—the whole 36-arm matrix reruns in under an hour, which is what made the three-generation lineage of Section VI-D affordable.

G. Generalization and zero-shot robustness

Following the constrained-DRL reporting standard [22], we evaluate the peak-trained policies *without retraining* along four test-time axes (Fig. 3; 5 points each, 50 rollouts per point, 3 seeds per learning arm) and at three unseen operating points (Fig. 4).

TABLE VIII: Sample efficiency: episodes to 90% of terminal smoothed return; 500-episode budget, 3 seeds.

Arm	ep-to-90%	budget
Proposed (peak)	3 ± 1	500
No-Lagrangian (peak)	2 ± 1	500
Fixed-penalty (peak)	3 ± 1	500
Legacy-LUT backend (peak)	2 ± 2	500
Flat PPO (peak)	27 ± 18	500
Quality-channel-off (peak)	3 ± 2	500
Warm-start-at-pin (peak)	2 ± 2	500
No-slow-head (peak)	3 ± 2	500
Proposed (nominal-trained)	146 ± 123	500

Fleet size (2–8 UAVs). Admitted mission is flat for every arm (proposed 0.172–0.191; escalation-aware oracle 0.909–0.916), and the arm ordering is preserved at every point. The padded observation and mask machinery transfer across fleet sizes without retraining.

Arrival load (12–28 queries per fixed burst horizon). The one axis with real motion, and it moves exactly as the admission-regime reading predicts: the learning arms decline smoothly (proposed 0.254 $\rightarrow$ 0.134) as the un-serveable fraction grows, while the escalation-aware oracle’s admitted mission *rises* (0.741 $\rightarrow$ 0.933) because its escalation rate climbs with overload (0.450 $\rightarrow$ 0.667)—the harder the domain, the more the certificate sheds and the cleaner the admitted core. The widening scissors between the oracle and the learning arms is the clearest picture of what strategy-layer learning does not solve at peak.

Link attenuation (+0 to +15.5 dB excess loss) and interference floor (−120 $\rightarrow$ −116 dBm). Both link axes leave admitted mission flat (proposed 0.180 $\rightarrow$ 0.174 and

0.180 $\rightarrow$ 0.179 respectively)—the extra attenuation drops the mean sensed SNR from 28.8 to 15.9 dB, still inside the calibrated quality plateau, and the raised noise floor shifts SINR by $\lesssim 2$ dB. At the peak operating point the bottleneck is the admission geometry, not the link: a useful robustness statement, and an honest one—these axes would bite only past the LUT support edge, where the outage guard (Section III) takes over.

Zero-shot triple. At the *unseen soft-conflict profile* (calibrated softer UTM geometry) the peak-trained constrained arms transfer well (proposed 0.619 ± 0.219 , fixed-penalty 0.714 ± 0.153) and beat the unconstrained arm (0.473)—and, notably, the escalation-aware oracle itself (0.486), whose certificate-triggered shedding is too aggressive for a near-feasible domain. At $1.5\times$ load the arms continue the arrival-axis trend (0.123). The *unseen low-SNR blockage profile* is the failure mode we highlight: the certificate marks essentially the whole critical load unserveable—the escalation-aware oracle escalates nearly everything and its admitted mission collapses to 0.000—while the zero-shot policies never engage escalation at all (critEsc 0.000) and keep serving with 0.22 admitted deadline violations (proposed 0.149 ± 0.085). The certificate transfers; the learned *shedding behaviour* does not. Making the escalate/serve split itself robust to unseen link regimes is the concrete open problem this sweep isolates.

H. Safety bridge, corrected: evidence choice as an airspace tax

Finally we close the loop between communication and separation with the M/G/1 coupling of Section III-D (Table IX, Fig. 5). At the -5 dB Rayleigh worst case with the WP5 2σ band, a token upload adds 7.5 ms of queueing wait to the tactical loop: $d_{TC} = 370.6$ m, indistinguishable from the BUBBLES default-comm baseline of 370.5 m, at 99.98% relative capacity. An image upload at peak shared-C2 load adds 1.98 s: $d_{TC} = 394.3$ m (+23.7 m over baseline) and 94.0% relative capacity ($\approx 6\%$ of airspace forfeited); under nominal load the tax is +11.2 m and $\approx 3\%$. A dedicated C2 band roughly halves the peak image tax (+17.2 m). *Honest correction:* an earlier version of this bridge substituted the payload airtime for the official 1.8 s separation-communication mean, so light evidence appeared to *shorten* d_{TC} below the deliverable’s baseline (-48.8 m, “ $1.14\times$ capacity”); that gain was an artifact of letting the payload stand in for the separation loop’s own exchange. Under the corrected queueing coupling nothing beats the baseline—heavy evidence taxes the airspace, light evidence avoids the tax—and the co-design signal survives in honest form: at low SNR, routing to tokens keeps the separation service at its certified baseline while the image path costs up to 6% of capacity precisely when spectrum is scarcest.

VII. DISCUSSION

A. Dual pricing is a domain property, not a method property

The central methodological lesson of Section VI-D is that the *same* Lagrangian machinery, unchanged, exhibits both textbook behaviours depending on where it operates. In the

TABLE IX: Corrected safety bridge at -5 dB Rayleigh, 2σ band, shared C2: M/G/1 wait W , contact distance, and relative capacity (baseline d_{TC} : 370.49 m; dedicated-C2 peak image in parentheses).

Evidence / load	W (s)	d_{TC} (m)	Δd_{TC} (m)	rel. capacity
s_1 token, nominal	0.004	370.53	+0.04	1.000
s_1 token, peak	0.008	370.58	+0.09	1.000
s_2 image, nominal	0.930	381.65	+11.2	0.971
s_2 image, peak	1.984	394.29	+23.8	0.940
(dedicated C2)	1.435	387.70	+17.2	0.956

near-feasible nominal domain the quality-critical multiplier climbs slowly and stays interior (3.29 at episode 500, never touching its cap of 8) against a realized cost that closes on its limit—a shadow price forming on the constraint boundary. In the peak overload domain the same multiplier ratchets monotonically to whatever ceiling it is given (~ 18.4 under a cap of 20 across two earlier generations; exactly 8.00 on 3/3 seeds under the v8 cap) against a cost stuck an order of magnitude above its limit. This is not a tuning failure: CMDP theory says a bounded interior multiplier exists only when the constraint set is feasible [16], [23]. The pin is the optimizer correctly reporting “no feasible policy exists here.”

Two experimental probes confirm the reading from opposite sides. The 150-episode dual warm-up lets the policy form before any constraint price moves, yet once ascent resumes the cost is still ≈ 0.37 and the climb recurs—shaping cannot manufacture a feasible region the environment does not contain. Conversely, warm-starting the multiplier at its pinned value (Section VI-E) changes nothing either: the trajectory’s destiny is set by the domain, not by its initial condition or its schedule.

B. In an infeasible domain, the cap is a behaviour knob—so use admission, not price

A subtler and, we believe, underreported effect: once the multiplier is pinned, its *ceiling* stops being a safety parameter and becomes a behaviour-shaping knob. Capping λ_{QC} at 8 instead of 20 halved the policy’s critical-escalation rate (0.126 $\rightarrow$ 0.086), dropping it *below* the certificate-derived budget band: at a pinned price of 18.4, escalation had been the cheap escape valve from the quality penalty; at 8 that pressure halves and the policy under-escalates. Every choice of cap trades one criterion against another because the dual no longer has an interior equilibrium to find. This coupling is exactly why constraint *pricing* is the wrong control surface for an overloaded safety system, and why our architecture hands that regime to the certified escalation/admission layer with an explicit budget (Section III-F)—the dedicated feasibility mechanism that reachability-constrained RL argues infeasible states require [24]. The escalation-aware oracle makes the point constructively: admitted-set mission success 0.916 is achievable at peak, but only by escalating 0.666 of the critical load—no strategy-layer price can be “paid” into a 0.155 escalation budget when the certificate says 0.105 of the load is structurally unserveable and the deadline geometry starves the rest.

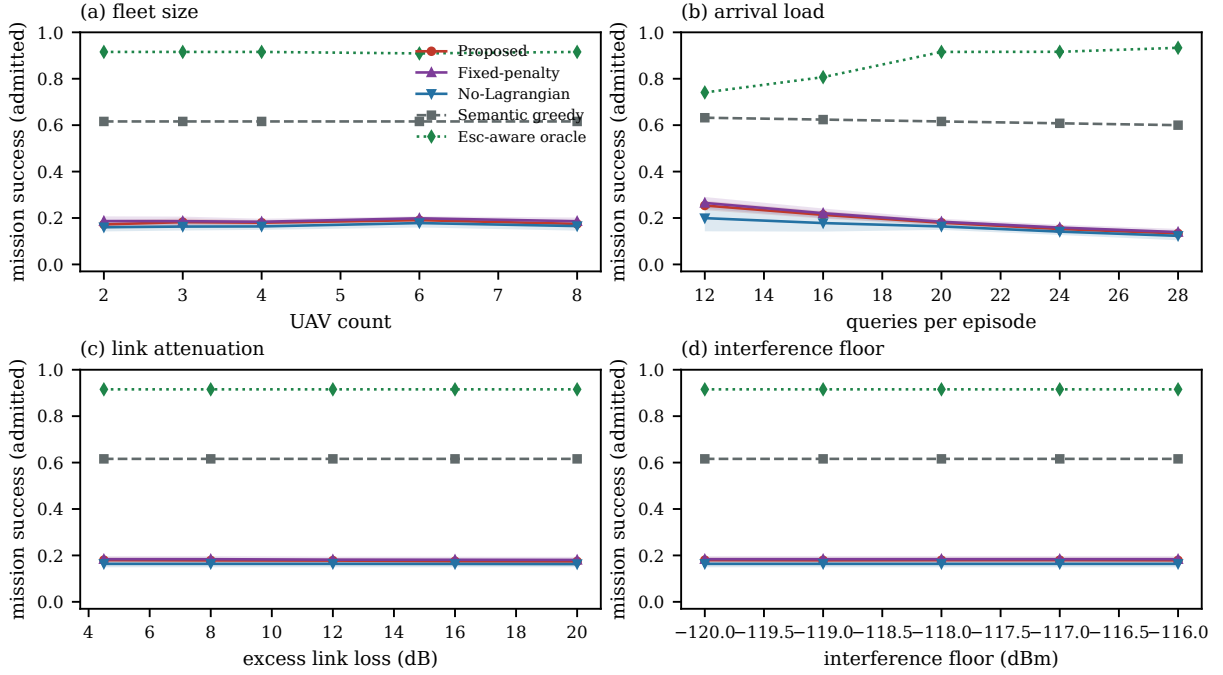


Fig. 3: Zero-shot generalization of the peak-trained policies along four axes (no retraining): (a) fleet size, (b) arrival load on the fixed burst horizon, (c) excess link attenuation, (d) interference floor. Bands: $\pm$ std over 3 seeds.

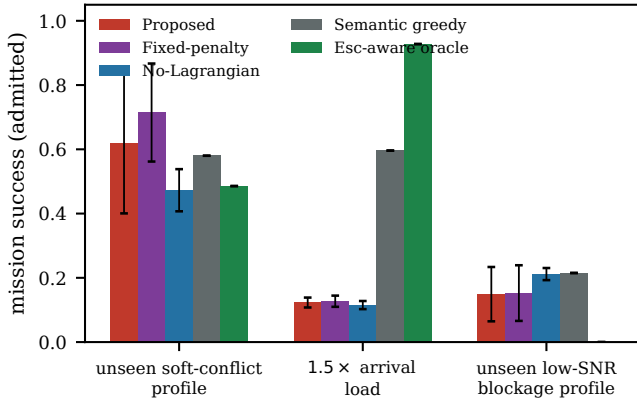


Fig. 4: Three-point zero-shot evaluation: unseen soft-conflict profile, $1.5\times$ arrival load, unseen low-SNR blockage profile.

C. The constrained arm equalling the unconstrained arm is a finding, not a defect

At peak, the proposed constrained policy and its unconstrained ablation converge to statistically indistinguishable behaviour. We report this plainly because the mechanism is now visible: a saturated-and-uniform dual penalty is a constant offset that cannot shape *relative* action values, so in the infeasible domain the dual term carries no gradient information. The reward-wiring fix of Section V-C is what makes this interpretable—after the fix, the *unconstrained* arm’s peak behaviour improved markedly (cache flooding $0.436 \rightarrow 0.298$, task success $0.242 \rightarrow 0.384$), proving the reward path is load-bearing, while the *constrained* arm’s peak behaviour stayed

invariant across three generations of reward and dual surgery (admitted mission $0.179/0.179/0.180$). The invariance isolates the failure to the pricing mechanism itself, not to plumbing. We spent three engineering generations (v6 scale-consistency gates, v7 link/reward realism, v8 dual levers) trying to make the peak dual price work before accepting the verdict; we document that lineage as the motivation for the admission-regime framing rather than presenting the architecture as designed-right-first-time.

D. Communication as a flight-safety variable, corrected

The safety bridge remains, but its sign discipline changed once the coupling was corrected from substitution to queueing (Sections III-D and VI-H): payload can only *add* waiting time to the tactical loop, so the honest headline is asymmetric—heavy evidence *taxes* the airspace ($+23.7$ m, -6% capacity at peak), light evidence *avoids the tax* ($+9$ cm), and nothing beats the BUBBLES baseline. The operational co-design signal survives in this honest form: at low SNR the scheduler should bias toward tokens not only for deadlines but to keep the separation service at its certified baseline, precisely when spectrum is scarcest. The claim is auditable at the parameter level against the deliverable, including the contested C2-spectrum assumption, which we bound by evaluating both shared and dedicated C2 modes.

E. System closure with the companion study

The companion paper [1] establishes, in isolation, that question-conditioned evidence routing beats full-image transmission—even over an error-free channel—and that a

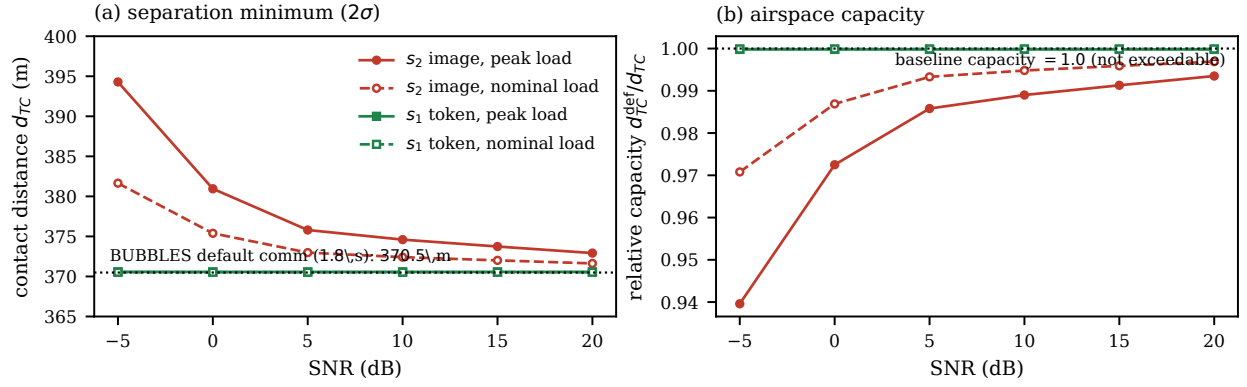


Fig. 5: Corrected safety bridge (Rayleigh, 2σ): (a) tactical-contact separation d_{TC} and (b) relative airspace capacity vs. SNR for token and image under peak/nominal shared-C2 load. The M/G/1 coupling adds queueing wait to the official 1.8 s mean, so no evidence level beats the BUBBLES baseline; images pay an airspace tax that grows as the link weakens.

zero-parameter rule matches a data-calibrated selector. The present paper embeds that per-question capability into a multi-UAV, safety-constrained scheduler and shows what survives contention and regulation: the same evidence levels that win on accuracy also keep the separation service at its baseline, and the scheduling question becomes one of admission under overload. Together the two studies form a closed loop from per-query evidence selection up to airspace-level resource and safety management. A federated extension—sharing evidence-selection statistics across UAVs and edge sites while keeping the safety kernel local—is the natural next step toward a deployable U-space semantic service.

F. Limitations

We keep the red lines explicit. *First*, the peak condition is reported as a stress test that exceeds the strategy layer’s feasible region: admitted-set mission success of the learned policy (0.180) sits far below the escalation-aware oracle (0.916), and we claim only that the safety accounting is carried structurally there—by the certificate and the budgeted escalation channel—not that the learned policy closes the gap. *Second*, the certificate and the quality constraints inherit the calibrated LUT quality model; a backend swap (legacy LUT) shifts absolute numbers (the dual pin persists, 2/2 seeds), so claims are conditional on the calibration. *Third*, the conflict-budget normalization constant (0.08) is an engineering mapping within the BUBBLES barrier-ladder framework, declared as a design choice rather than a value read from the deliverable; the same holds for the M/G/1 offered-load presets (ρ values) behind Table I. *Fourth*, the escalation channel’s budget is calibrated from the certificate roll, but the 0.05 headroom is a design margin; under the v8 cap the learned policy under-escalates (0.086 vs. the [0.105, 0.205] band), and the zero-shot blockage profile shows the learned shedding behaviour does not transfer to unseen deeply-infeasible link regimes (Section VI-G)—both reported as open behaviour-shaping issues of pinned-dual regimes rather than hidden. *Fifth*, per-seed variance is elevated by the no-GAE Monte-Carlo advantage estimate, a deliberate simplicity trade for the short dense-reward episodes.

Finally, we do *not* claim a certified system: the framework is BUBBLES-conformant at the parameter level, and all safety statements are simulation statements about that model, not about flight-tested hardware.

VIII. CONCLUSION

We presented a U-space/BUBBLES-conformant framework for jointly scheduling semantic evidence levels and airspace resources in multi-UAV visual question answering, in which every scenario parameter is traceable to a SESAR JU deliverable and the separation kernel reproduces the official numerics to within 0.18%. A corrected safety bridge couples the payload to the tactical-contact separation chain through an M/G/1 priority queue: tokens keep the separation service at its certified baseline, while peak image traffic inflates the contact distance by 23.7 m and forfeits $\approx 6\%$ of relative airspace capacity—communication remains a flight-safety variable, now with honest sign discipline. Scheduling is formalized as a CMDP on the admitted set with a certificate-based, budget-calibrated escalation channel, and solved by a risk-aware constrained hybrid-action PPO behind a six-layer constraint stack. The central empirical result is a dual-pathology contrast obtained by running the same machinery in a feasible and an infeasible regime: under nominal load the quality dual converges to an interior shadow price (3.29) and the policy attains 93% of the escalation-aware oracle’s admitted mission success; under peak overload the dual ratchets to its cap on 3/3 seeds, the constrained and unconstrained arms coincide, and safety is carried structurally—the escalation-aware oracle reaches 0.916 only by escalating 0.666 of the critical load, while the conflict, critical- deadline, and escalation budgets all hold with margin. The methodological takeaway—read a pinned multiplier as an infeasibility certificate and reach for admission control rather than a higher price, verifying with channel-off and warm-start-at-the-pin ablations—is the paper’s most portable contribution. Future work will close the nominal quality gap that the interior dual is still pricing at episode 500, learn the escalate/serve split directly against the certificate budget, and pursue a federated extension that shares evidence-selection statistics across a U-space fleet while keeping the safety kernel local.

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